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1 Introduction

This document provides usage information for Time Manager.

The Time Manager is a service that provides the system with support for handling multiple UTC time sources (e.g. GNSS).

This document will cover details about features provided by Time Manager, configuration usage, and dynamic behavior. The target audience for this document are developers and architects who will use the Time Manager service.

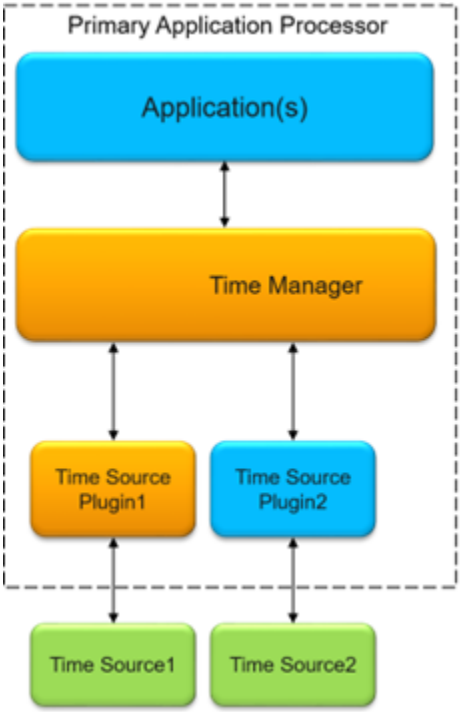

Figure 1. Time Manager Block Diagram

1.1 Notational Conventions

Ctrl+x	*A sequence with a '+' indicates that you must press and hold one key or pointing device button while pressing another key or pointing device button*
[ ]	*Optional elements in a command*
…..	*Indicates one of the following possibilities:* • *Additional optional arguments in a statement have been omitted* • *The preceding item or items can be repeated one or more times* • *Additional parameters, values, or other information can be entered*
-	*Line continuation: indicates that a command line or code line continues on the following line when at the end of a line*
Italic text	*Indicates important information, complete titles of manuals or variables*
`Mono-spaced`	*Example code or interactive screen display/output*
Numbers	*All numbers are assumed to be decimal unless otherwise noted. Non decimal radixes – binary, octal or hexadecimal – are explicitly indicated*
Underline blue text	*A link to an external reference (e.g., website, document)*

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`Modules`
`|-> section`
`|-> item`
	A link to a section or item in the ConMod TP html interface specification (Doxygen output from platform code headers)

Diagram Legend

- Telematics Platform
- Product Specific
- 3rd Party

2 Setup

2.1 Tools/Equipment Needed

Not Applicable.

2.2 Setup Instructions

How To Be a Client of Time Manager

1) Use pkg-config in the client application Makefile to link to the Time Manager client library

```
LDFLAGS += `pkg-config --libs libcas-time-mgr`
CFLAGS += `pkg-config --cflags libcas-time-mgr`
```

2) Include Time Manager header file in client application source code

```
#include "timemgr.h"
```

3) Wait for Time Manager to be ready at startup

   a. If client application has a “hard” dependency on Time Manager to be running use systemd .service “After” field

```
[Unit]
After=timemgr.service
```

   b. If client application has only a single thread (or portion of process) that is dependent on Time Manager use TP service monitor API (see TP_F101U-00023 for details):

```
tpSVCMON_registerServiceStateChange(TM_SERVICE, FALSE,
callback, &monitor_id);
```

4) **Security:** Usage of Time Manager APIs require clients (user) to be a member of **“time”** group.

5) Once Time Manager service is ready, initialize communication with the Time Manager in client application source code (N**ote:** pre-requisite is that the client must first call `tpSYS_initIPC()`)

```
tpTM_Error_t tm_ret = E_TM_ERROR_NONE;
tm_ret = tpTM_init();
/* check return code */
```


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6) To de-initialize the Time Manager in client application source code

```
tm_ret = tpTM_deinit();
```

Note: The application must ensure to call `tpTM_deinit()` once interaction with the Time Manager are finalized; this is needed to ensure that timers are released.

Time Manager configuration

`timemgr_config.xml` file specifies the persistent data that Time Manager requires for operation. ConMod TP SW provides a default Time Manager configuration.

By default the following configuration is provided:

```
<PLATFORM_CONFIG>

        <service name="Time Manager">
                <property name="time_threshold" value="5"/>
                <plugins vendor="Continental">
                        <plugin name="rtc_gnss">
                                <property name="type">source</property>
                        </plugin>
                </plugins>
        </service>
</PLATFORM_CONFIG>
```

Note: Time manager configuration described upon it is just for reference, it might differ from the actual implementation.

2.3 Setup Troubleshooting

Not Applicable.

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3 Features/Components

3.1 Time Sources

3.1.1 Overview

3.1.1.1 Platform Content Overview

Time Manager integrates all supported time sources into a single point. The data provided by the different time sources can be used for applications purposes. This feature allows Time Manager clients to retrieve the time from internal and external time sources.

ConMod TP SW provides plugin for RTC_GNSS time source.

Time reported by platform is in Coordinated Universal Time (UTC) format. For further details about UTC, refer to https://en.wikipedia.org/wiki/Coordinated_Universal_Time. Time Manager handles a resolution of nanoseconds.

3.1.1.2 Product Expectations

Platform is configured properly (see Time Manager configuration section for details).

3.1.2 Setup

3.1.2.1 Tools/Equipment Needed

Not Applicable.

3.1.2.2 Setup Instructions

Not Applicable.

3.1.2.3 Setup Troubleshooting

Not Applicable.

3.1.3 Usage

3.1.3.1 Basic Instructions

Not Applicable.

3.1.3.2 Detailed Instructions

References to Time Manager APIs and types that can be found in this section, for more details on these APIs and types see the Doxygen links in the following table:

API/Type Description	Doxygen Path

Callback function on new query time event reception.	`Modules`
	`  |-> Housekeeping Infrastructure`
	`  |-> Time Manager (TM)`
	`  |-> tpTM_QueryTimeEventCallback_t`

API to register a query time event callback function.	`Modules`
	`  |-> Housekeeping Infrastructure`
	`  |-> Time Manager (TM)`
	`  |-> tpTM_registerQueryEventCallback()`

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API/Type Description	Doxygen Path
API to deregister a query time event callback function.	Modules |-> Housekeeping Infrastructure |-> Time Manager (TM) |-> **tpTM_deregisterQueryEventCallback()**
API to start time query from a specific source.	Modules |-> Housekeeping Infrastructure |-> Time Manager (TM) |-> **tpTM_startQuery()**
API to stop a time query from a specific source.	Modules |-> Housekeeping Infrastructure |-> Time Manager (TM) |-> **tpTM_stopQuery()**
API to get time query state from a specific source.	Modules |-> Housekeeping Infrastructure |-> Time Manager (TM) |-> **tpTM_getQueryState()**
API to request the time from a specific time source.	Modules |-> Housekeeping Infrastructure |-> Time Manager (TM) |-> **tpTM_getSourceTime()**
API to query the list of input sources used by Time Manager.	Modules |-> Housekeeping Infrastructure |-> Time Manager (TM) |-> **tpTM_getSourceList()**
API to get the detailed time information from the Time Source.	Modules |-> Housekeeping Infrastructure |-> Time Manager (TM) |-> **tpTM_getDetailedSourceTimeInfo**

Time Manager enables the support of Asynchronous and Synchronous time sources. Asynchronous time sources notify time primarily via events. Meanwhile, synchronous time sources require a request to retrieve and return the time.

Use Case: Get time from an specific time source

Time Manager provides the ability for an application to request the time from a specific source. The application should call *`tpTM_getSourceTime()`* API with a *`sourceName`* argument corresponding to a valid time source used by Time Manager. Time Manager will retrieve the last known time from the time source plugin, **not the Linux system time.**

ConMod TP SW provides setup and plugin for RTC_GNSS time source only.

`tpTM_getSourceList()` API can be used to obtain the list of time sources that are configured to be used by Time Manager. Such list is derived from the time source plugins that are configured via `timemgr_config.xml` file.

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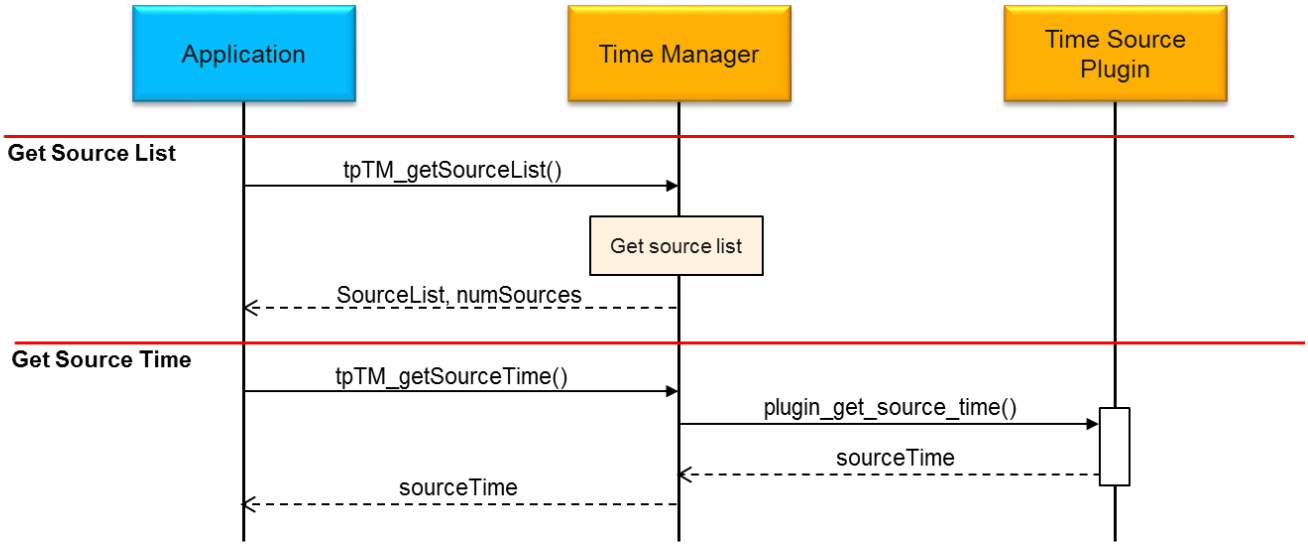

Figure 2. Get Source Time

Use Case: Query time from non-periodically reported time source

Time Manager supports the specific use case where the Time Manager client has the need to retrieve the time from a synchronous time source; these types of time sources will not periodically report time.

Time Manager does not provide the means to differentiate between synchronous or asynchronous time sources. System Architect(s) need to have a clear understanding of the supported time sources in order to identify the correct procedure to obtain the time from a specific time source, either synchronously or asynchronously.

The application should call *`tpTM_registerQueryEventCallback()`* API before starting a query from a specific source, this in order to be able to be notified when a query event is received. Only after the previous step, the *`tpTM_startQuery()`* API should be called to start the query. Once a query is started, it can be stopped by calling *`tpTM_stopQuery()`* API.

> **Note:** Only one outstanding query per time source is allowed by the Time Manager. If more than one Time Manager client is expected to start queries from a time source, *`tpTM_getQueryState`*() along with *`tpTM_registerQueryEventCallback()`* API can be used to verify when a new query can be started.

The application will receive asynchronous events reporting *`tpTM_QueryTimeEventCallback_t`* each time a change in the query state is detected (see Figure 3). The following query information will be provided:

- *`tpTM_QueryState_t`*: Either `idle`, `in progress`, or `complete`
- *`tpTM_QueryStatus_t`*: Either `none`, `retry`, `failure`, or `success`
- *`tpTM_Time_t`*: time reported on success; **Time only relevant if status = success.**

> **Note:** The time reported by *`tpTM_QueryTimeEventCallback_t`* will correspond to the last know time reported by the time source. If the time source has not yet reported a time, the value returned will be zero

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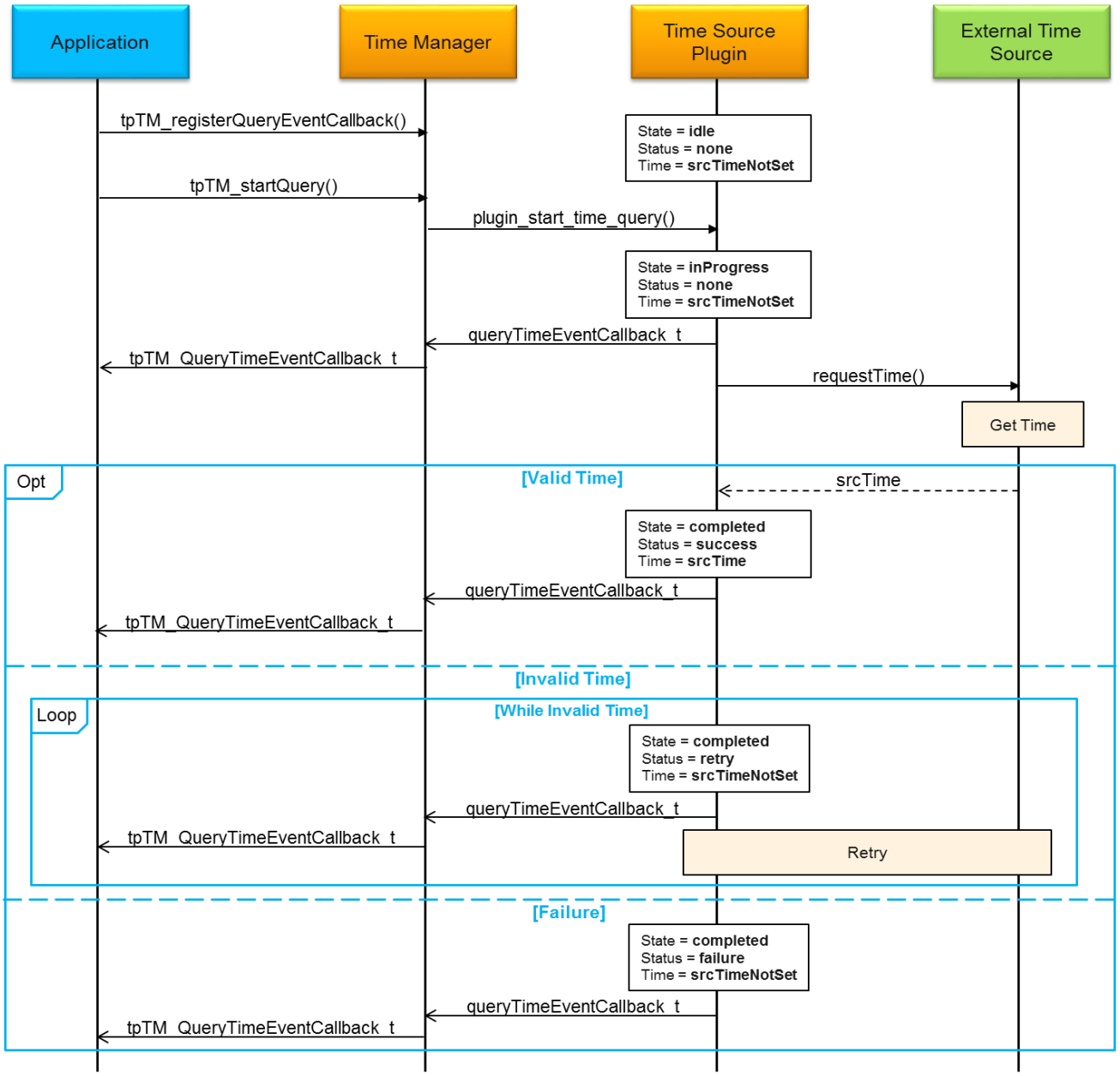

Figure 3. Query time events notification

Use Case: Get state of a query

Time Manager provides the ability for an application to request the state of a time source query synchronously when using `tpTM_getQueryState()` API.

The Time Manager will return the queries' last known state, status and source time (see Use Case:Query time from non-periodically reported time source for more details).

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3.1.3.3 Testing/Error Reporting

For an extensive list of error codes, see *tpTM_Error_t* type definition. The following table explains some of these error codes in the context of Time Source query.

Error Condition	Possible cause	Corrective Action
`tpTM_startQuery()` API returns `E_TM_ERROR_UNSUPPORTED`	The time source that is being queried does not support querying mechanism.	Use `tpTM_getSourceTime()` to retrieve the source time.
`tpTM_getSourceTime()` API returns `E_TM_ERROR_NO_PLUG_FUNC`	Source name provided to `tpTM_getSourceTime()` does not match any of the configure time sources.	Ensure source name is defined, use `tpTM_getSourceList()` to obtain the list of supported time sources.
When trying to call any query related API, `E_TM_ERROR_NO_PLUG_FUNC` is returned	Source name provided to API does not match any of the configure time sources.	Ensure source name is defined, use `tpTM_getSourceList()` to obtain the list of supported time sources.
`tpTM_startQuery()` API returns `E_TM_ERROR_INTERNAL`	The time source is already processing a query request.	Wait for the ongoing query to finish, and start the new query after it.
`tpTM_registerQueryEventCallback()` API returns `E_TM_ERROR_INVALID_ARGUMENT`	Callback function pointer provided is NULL or invalid address.	Ensure callback function pointer points to a valid callback function of type `tpTM_QueryTimeEventCallback_t`.
When trying to call any time source related API, `E_TM_ERROR_DL_ERROR` is returned	The source plugin not loaded.	1. Ensure that the plugin name is correct. 2. Verify that Time Manager is properly configured (XML file, see Time Manager configuration section) 3. Ensure that the time source plugin library exist and it is not corrupt.

4 Known Issues

Not Applicable.

5 FAQs

Question:

Difference between “Time Source Event” and “Query Time Event”?

Answer:

Within Time Manager two types of time sources can be identified; the ones that support periodical transmission of time events and the time sources that need to be queried in order to retrieve its time.

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Time source events are those notification that are automatically passed to the Time Manager by the Time Source Plugins. These events are generated periodically based on the time source configuration e.g. GNSS and Location.

Query time events are events which are generated only after a query to a specific time source has been started by calling the `tpTM_startQuery()` API. An event will be notified each time a change in the query state is detected, and will stop once the query is stopped or completed e.g. NTP

6 Troubleshooting/Tips and Tricks

Not Applicable.

7 References

7.1 Abbreviations/Acronyms/Definitions

TM	Time Manager

7.2 Platform Documentation

TP_102G-00001: Telematics Platform Glossary

TP_F101U-00023: User Guide Common Library

7.3 TP_F101U-00026: User Guide PersistenceWebsite links

- Network Time Protocol project home page: http://www.ntp.org/
- UTC information Wikipedia page: https://en.wikipedia.org/wiki/Coordinated_Universal_Time

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8 Revision History

Version	Date	Description
01.00	May 25, 2024	3.1 Time Sources for ConMod. Initial release

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